A Notation list.

Table 4 describes the important notations used in our work.

Table 4: Summary of Notations.

Notation	Definition
t	The current task ID
$\mathcal{R}_{1:t} \triangleq \bigcup_{i=1}^{t-1} \mathcal{R}_i \cup \mathcal{R}_t$	The real data for tasks from 1 to $t-1$
$\mathcal{R}_t \triangleq \{(x_i, y_i)\}_{i=1}^{n_t}$	The real data for task-t
x_i	The real data sample
y_i	The label of the real data sample x_i
n_t	The number of real instances in task-t
C_t	The whole label space for task-t
$\mathcal{H}_{1:t-1} \triangleq \bigcup_{i=1}^{t-1} \mathcal{H}_i$	The hybrid memory for tasks from 1 to $t-1$
$\mathcal{H}_t \triangleq \{\mathcal{S}_t, \mathcal{R}_t[A_t]\}$	The hybrid memory for the task-t
$\mathcal{S}_t$	The synthetic exemplars for the task-t
$\mathcal{S}_{t,j}$	The updated synthetic exemplars for the task-t after epoch-j of task-t in sliding window version of CDD
τ	The window size (the number of cached checkpoints) for sliding window version of CDD
η	The learning rate for updating the synthetic exemplars $\mathcal{S}_{t,j}$ by sliding window version of CDD
$A_t \subseteq \{1, 2, \cdots, \lvert\mathcal{R}_t\rvert\}$	The indexes of subset chosen from the real dataset $\mathcal{R}_t$
$f(\cdot;\theta)$	The whole model, with parameters θ
$\theta_{\mathcal{H}_t}$	The optimal model trained on task-t's hybrid memory
$\theta_{\mathcal{R}_t}$	The optimal model trained on task-t's real data
$\theta_{\mathcal{R}_{1:t}}$	The optimal model trained on the real data of all tasks from 1 to t
$\theta_{\mathcal{H}_{1:t}}$	The optimal model trained on the hybrid memory of all prior tasks from 1 to t
ϵ_t	The scalar value bounding the performance of $\theta_{\mathcal{H}_{1:t}}$ on $\mathcal{R}_{1:t}$ by the performance of $\theta_{\mathcal{R}_{1:t}}$ on $\mathcal{R}_{1:t}$.
ρ	The scaling factor that bounds the sum of the performance of $\theta_{\mathcal{H}_{1:t}}$ on $\mathcal{R}_{1:t}$ and the performance of $\theta_{\mathcal{H}_{t+1}}$ on $\mathcal{R}_{t+1}$.

B Algorithm of Conditional Real Data Selection

As described in Sec. 4.3, we employ a greedy algorithm for our proposed conditional real data selection. The details of this algorithm are provided below,

Algorithm 2: Conditional Real Data Selection (Greedy Algorithm)

input : Real data for classes C_t: $\mathcal{R}_t$; synthetic exemplars: $\mathcal{S}_t$; number of selected exemplars per class: k
initialize : Selected subset $A_t \leftarrow \emptyset$, minimum distance $d_{min} \leftarrow +\infty$

```
while |A_t| < |C_t| × k do
    for i ∈ V_t \ A_t do
        A'_t ← A_t ∪ {i};
        compute the objective value in Eq. 4 as d, where A_t is set to A'_t;
        if d < d_min and |{j ∈ A_t : y_j = y_i}| < k then
            Update minimum distance: d_min ← d;
            i* ← i;
    Update subset A_t ← A_t ∪ {i*};
```

output : Selected exemplars $\mathcal{R}_t[A_t]$

C The Proof For Theorem 1.

Here we provide the detailed proof for Theorem 1.

Theorem 1 (Performance Approximation)**.** Based on the above Assumptions 1 and 2, when $\epsilon_{t+1} \geq \frac{\rho}{1-\epsilon_t}$, we can derive that the model trained on the hybrid memory of all previous tasks achieves performance comparable to that of the model trained on the real dataset of all previous tasks,

$$\log P(\mathcal{R}_{1:t}|\theta_{\mathcal{H}_{1:t}}) \geq (1-\epsilon_t)\log P(\mathcal{R}_{1:t}|\theta_{\mathcal{R}_{1:t}}). \tag{6}$$

Proof. According to the Definition 4, the $\theta_{\mathcal{R}_{1:t}}$ is the optimal solution of $r_t(\theta)$, thus we can get $r_t(\theta_{\mathcal{R}_{1:t}}) \geq r_t(\theta_{\mathcal{R}_{1:t+1}})$. And according to the Definition 2, the $\theta_{\mathcal{R}_{t+1}}$ is the optimal solution of $\log P(\mathcal{R}_{t+1}|\theta)$, thus we can get $\log P(\mathcal{R}_{t+1}|\theta_{\mathcal{R}_{t+1}}) \geq \log P(\mathcal{R}_{t+1}|\theta_{\mathcal{R}_{1:t+1}})$.

According to the Mathematical Induction:

Base Case ($t = 1$):

$$\log P(\mathcal{R}_{1:1}|\theta_{\mathcal{H}_{1:1}}) = \log P(\mathcal{R}_1|\theta_{\mathcal{H}_1}),\ \ \log P(\mathcal{R}_{1:1}|\theta_{\mathcal{R}_{1:1}}) = \log P(\mathcal{R}_1|\theta_{\mathcal{R}_1})$$

$$\exists\epsilon \in [0,1),\ \ \log P(\mathcal{R}_1|\theta_{\mathcal{H}_1}) \geq (1-\epsilon)\log P(\mathcal{R}_1|\theta_{\mathcal{R}_1})$$

According to the Assumption 1, the base case holds true.

Inductive Hypothesis: Assume that the formula holds for ϵ_t at task-t:

$$\log P(\mathcal{R}_{1:t}|\theta_{\mathcal{H}_{1:t}}) \geq (1-\epsilon_t)\log P(\mathcal{R}_{1:t}|\theta_{\mathcal{R}_{1:t}})$$

Inductive Step: We will prove that the formula holds for ϵ_{t+1} at task-$t+1$:

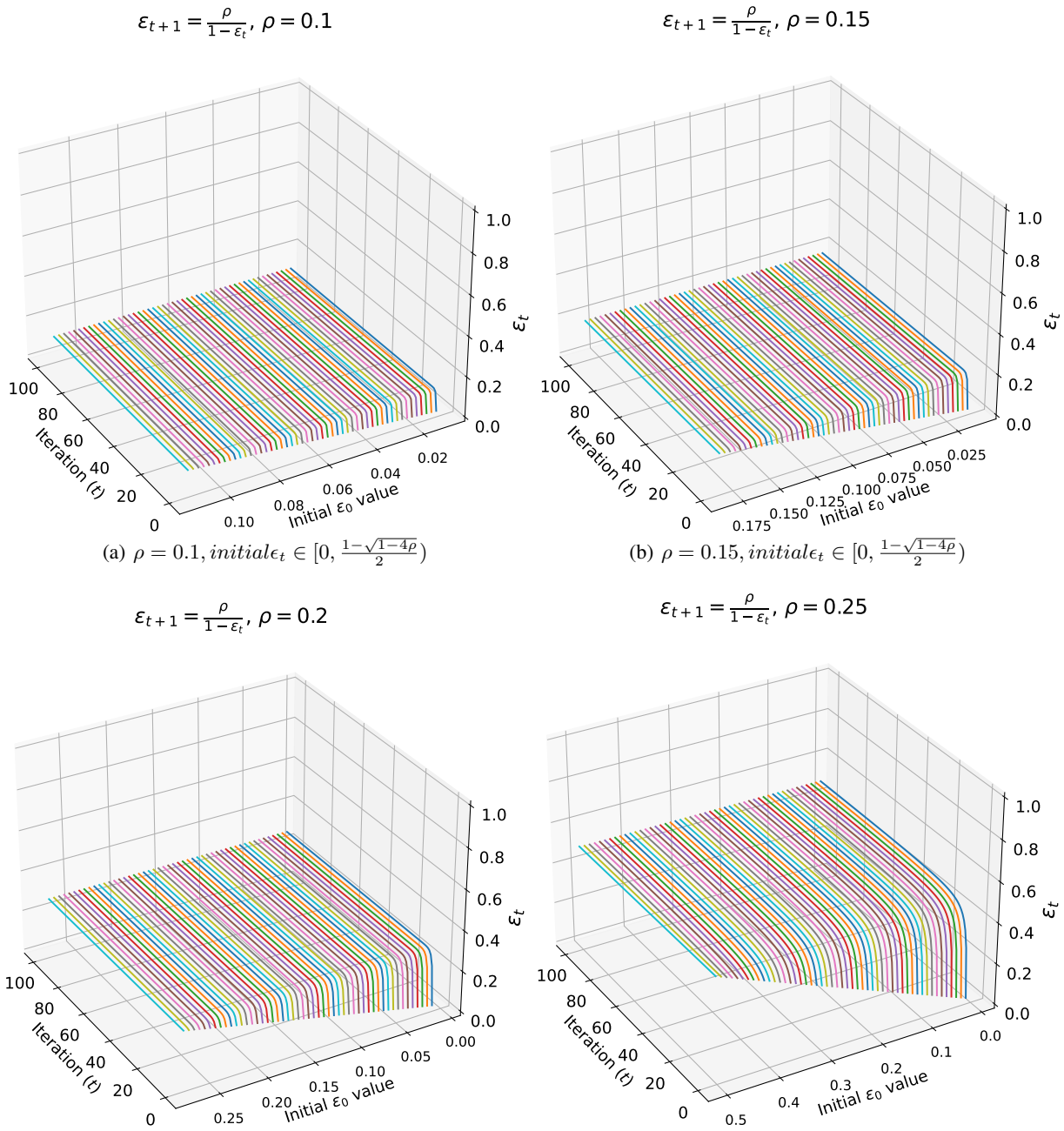
$$\begin{aligned}
\log P(\mathcal{R}_{1:t+1}|\theta_{\mathcal{H}_{1:t+1}}) &= \log P(\mathcal{R}_{1:t}|\theta_{\mathcal{H}_{1:t+1}}) + \log P(\mathcal{R}_{t+1}|\theta_{\mathcal{H}_{1:t+1}}) \\
&= r_t(\theta_{\mathcal{H}_{1:t+1}}) + \log P(\mathcal{R}_{t+1}|\theta_{\mathcal{H}_{1:t+1}}) \\
&= r_{t+1}(\theta_{\mathcal{H}_{1:t+1}})
\end{aligned}$$

$$\begin{aligned}
\log P(\mathcal{R}_{1:t+1}|\theta_{\mathcal{R}_{1:t+1}}) &= \log P(\mathcal{R}_{1:t}|\theta_{\mathcal{R}_{1:t+1}}) + \log P(\mathcal{R}_{t+1}|\theta_{\mathcal{R}_{1:t+1}}) \\
&= r_t(\theta_{\mathcal{R}_{1:t+1}}) + \log P(\mathcal{R}_{t+1}|\theta_{\mathcal{R}_{1:t+1}}) \\
&= r_{t+1}(\theta_{\mathcal{R}_{1:t+1}})
\end{aligned}$$

$$\begin{aligned}
r_{t+1}(\theta_{\mathcal{R}_{1:t+1}}) - r_{t+1}(\theta_{\mathcal{H}_{1:t+1}}) &\leq r_t(\theta_{\mathcal{R}_{1:t+1}}) + \log P(\mathcal{R}_{t+1}|\theta_{\mathcal{R}_{1:t+1}}) \\
&\leq r_t(\theta_{\mathcal{R}_{1:t}}) + \log P(\mathcal{R}_{t+1}|\theta_{\mathcal{R}_{t+1}}) \\
&\leq \frac{1}{1-\epsilon_t}\left(r_t(\theta_{\mathcal{H}_{1:t}}) + \log P(\mathcal{R}_{t+1}|\theta_{\mathcal{H}_{t+1}})\right) \\
&\leq \frac{\rho}{1-\epsilon_t} r_{t+1}(\theta_{\mathcal{R}_{1:t+1}}) \\
&\leq \epsilon_{t+1} r_{t+1}(\theta_{\mathcal{R}_{1:t+1}})
\end{aligned}$$

Thus, by the principle of mathematical induction, the formula holds for all ϵ_t of task-t. □

Discussion. We discuss the performance boundary of the model trained with the proposed hybrid memory. According to Theorem 1, the performance of the optimal model trained on the hybrid memory for all tasks which is denoted as $r_t(\theta^H_{1:t})$, is bounded by the performance of the optimal model trained on the real dataset for all tasks, scaled by $1-\epsilon_t$, when $\epsilon_{t+1} \geq \frac{\rho}{1-\epsilon_t}$. It is evident that ϵ_t can converge to $\epsilon_t = \frac{1\pm\sqrt{1-4\rho}}{2}$ if $\rho \leq \frac{1}{4}$, ensuring that ϵ_t remains within the bounds $\epsilon_t \in [0,1)$. Therefore, if the initial $\epsilon_t \leq \frac{1\pm\sqrt{1-4\rho}}{2}$, the performance will be well bounded. We also illustrate how ϵ_t changes iteratively over 100 tasks with different values of ρ and initial ϵ_t in Fig. 6.



(a) $\rho = 0.1, initial \epsilon_t \in [0, \frac{1-\sqrt{1-4\rho}}{2})$ (b) $\rho = 0.15, initial \epsilon_t \in [0, \frac{1-\sqrt{1-4\rho}}{2})$

(c) $\rho = 0.2, initial \epsilon_t \in [0, \frac{1-\sqrt{1-4\rho}}{2})$ (d) $\rho = 0.25, initial \epsilon_t \in [0, \frac{1-\sqrt{1-4\rho}}{2})$

Figure 6: Visualization of ϵ_t in 100 iteration with different ρ and different initial ϵ_t.

D CDD Optimization Objectives.

We present a detailed description of several optimization objectives for our CDD approach.

CDD with DM. First, we describe the adaptation of DM [40] as the optimization objective for CDD, as outlined below,

$$\mathop{\mathbb{E}}_{\theta \sim \theta_{1:N}} [\ell_{DM}(\mathcal{S}_t, \mathcal{R}_t; \theta)] = \mathop{\mathbb{E}}_{\theta \sim \theta_{1:N}} \left[\| \mathbb{E}[\psi(\mathcal{S}_t; \theta)] - \mathbb{E}[\psi(\mathcal{R}_t; \theta)] \|^2 \right], \tag{8}$$

where N in Eq. 8 represents the total number of cached checkpoints θ in the history of learning task-t, $\psi(\cdot; \theta)$ is the function to extract the embedding of the input samples. The objective of Eq. 8 aims to measure the distance between the mean vector of the synthetic exemplars and the mean vector of the real dataset.